

Project Title

No-Code SaaS Data Ingestion and Analysis Using Amazon AppFlow, S3, and Athena

Problem Statement

A digital marketing agency collects customer interaction data from various SaaS platforms and wants to analyze this data without writing custom ETL pipelines. The goal is to:

- Ingest SaaS data with minimal code and effort
 - Store the data in Amazon S3
 - Query and analyze the data using Amazon Athena
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Objective

Set up a **no-code/low-code pipeline** using **Amazon AppFlow** to automatically collect data from a SaaS source, store it in **Amazon S3**, and analyze it using **Amazon Athena**.

AWS Services Used

Service	Purpose
Amazon AppFlow	To connect with SaaS platform and automate CSV data export to S3
Amazon S3	To store the exported CSV data from AppFlow
AWS Glue Data Catalog	To catalog and structure the data for querying in Athena
Amazon Athena	To run SQL queries on CSV files stored in S3
IAM	To manage permissions for Athena and the user to access S3

Sample Data Used

A simple CSV with customer details:

```
name,email,age
Alice,alice@example.com,28
```

Bob,bob@example.com,35
Charlie,charlie@example.com,42
Diana,diana@example.com,31
Ethan,ethan@example.com,27

Project Workflow

◇ Step 1: Prepare SaaS Data (Mock or Real)

- Choose a sample SaaS source (mock CSV file used in our case)
- Ensure CSV is formatted correctly

◇ Step 2: Set Up AppFlow

1. Go to Amazon AppFlow
2. Click **Create flow**
3. Choose source: e.g., **Amazon S3 (CSV upload simulating SaaS)**
4. Choose destination: **Amazon S3** (separate bucket/folder)
5. Set flow trigger: **On-demand**
6. Map source fields to destination

◇ Step 3: Run the Flow

- Upload the CSV file into AppFlow source folder
- Run the flow to push the file to destination S3 bucket

◇ Step 4: Check S3

- Confirm the CSV file is stored under: `s3://appflow-destination-bucket-saas/<auto-folder-path>/`

◇ Step 5: Create Athena Database & Table

```
CREATE DATABASE IF NOT EXISTS saas_db;
```

```
CREATE EXTERNAL TABLE IF NOT EXISTS saas_db.customers_csv (  
    name string,  
    email string,  
    age int  
)
```

```
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
```

```
WITH SERDEPROPERTIES (  
    "separatorChar" = ",",  
    "quoteChar" = "\""
```

```
)  
LOCATION 's3://appflow-destination-bucket-saas/<actual-folder-path>/'  
TBLPROPERTIES ('skip.header.line.count'='1');
```

Replace <actual-folder-path> with your real AppFlow output path.

◇ Step 6: Query the Data Using Athena

```
SELECT * FROM saas_db.customers_csv LIMIT 10;
```

IAM & S3 Permissions Setup

✓ Bucket Policy on S3 to allow Athena read access:

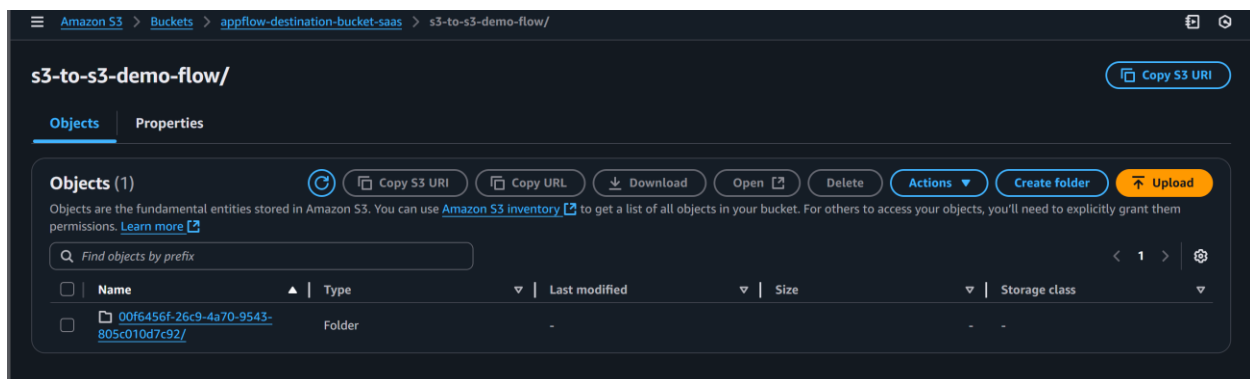
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowAthenaAccess",
      "Effect": "Allow",
      "Principal": "*",
      "Action": ["s3:GetObject", "s3:ListBucket"],
      "Resource": [
        "arn:aws:s3:::appflow-destination-bucket-saas",
        "arn:aws:s3:::appflow-destination-bucket-saas/*"
      ]
    }
  ]
}
```

✓ IAM user (data_engineer) also has inline policy for S3 access.

Output Screenshot (Athena Query)

 Paste a screenshot here of your Athena query result.

[S3 Output Screenshot]



[Athena Query Result Screenshot]

The screenshot displays the Amazon Athena console interface. On the left, the 'Data' sidebar shows the 'Data source' as 'AwsDataCatalog', 'Catalog' as 'None', and 'Database' as 'saas_db'. Under 'Tables and views', the 'customers_csv' table is listed with columns: name (string), email (string), and age (int). The main panel shows 'Query 1' with the SQL statement: `SELECT * FROM saas_db.customers_csv LIMIT 10;`. Below the query, the 'Query results' tab is active, showing a 'Completed' status with a green checkmark. The 'Query stats' bar indicates: 'Time in queue: 59 ms', 'Run time: 415 ms', and 'Data scanned: -'. The 'Results (0)' section shows a table with columns: #, name, email, and age. The footer includes 'CloudShell', 'Feedback', and copyright information for Amazon Web Services.

[AppFlow Run Screenshot]

The screenshot shows the Amazon AppFlow console. The left sidebar contains navigation links: Connectors, Flows, Connections, Users, and Documentation. The main area displays details for a flow named 's3-to-s3-demo-flow'. The flow is 'Active' and has a status of 'On demand'. The 'Flow description' is empty. The 'AWS KMS key' is 'AWS managed key'. The 'Date created' is 'June 24, 2025, 14:29'. The 'Source name' is 'Amazon S3', the 'Source bucket' is 's3://appflow-source-bucket-saas/company.csv', and the 'Destination name' is 'Amazon S3'. The 'Destination bucket' is 's3://appflow-destination-bucket-saas/s3-to-s3-demo-flow/'. The 'Trigger' is 'On demand'. The 'Last run' is 'June 24, 2025, 14:34'. The 'Schema version' is '1'. The 'Run metrics' link is 'CloudWatch metrics'. Below this, the 'File format settings' section shows 'Written data format' as 'JSON format' and 'File name format' as 'No timestamp'. The 'Data field settings' tab is active, showing 'Field mappings (1)'. The 'Field mappings' table has columns: Source field name, Destination field name, Formula, and Data modifications. The mapping shows 'name,email,age' mapped to 'name,email,age' with a formula of '-' and data modifications of '-'. The footer includes the AWS logo, 'Services', a search bar, and the user 'data_engineer @ 5350-0286-4770'.

Key Learnings

- Amazon AppFlow simplifies SaaS to S3 data ingestion without custom code

- Athena + Glue Data Catalog enables serverless querying over raw CSV files
 - Proper IAM and bucket policies are crucial for access control
 - This architecture can be scaled to integrate with real SaaS services like Salesforce, Zendesk, Google Analytics, etc.
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Project Tags

AWS Amazon AppFlow Amazon S3 Athena Glue Catalog Serverless ETL Data Engineering

Date & Name

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